SeedingMinds

Assessment Name

Java Coding_S1_170126

Module: Coding

Question 1:

Marks: +10 -0

Cyclic Array Rotation

You are given an array of `n` integers. Your task is to rotate the array to the right by `k` steps. A right rotation by one step means the last element of the array moves to the first position, and all other elements shift one position to the right.

For example, if the array is `[1, 2, 3, 4, 5]` and `k = 2`, the array should become `[4, 5, 1, 2, 3]`.

Input Format:

- The first line contains two space-separated integers, `n` (the number of elements in the array) and `k` (the number of rotations).
- The second line contains `n` space-separated integers, representing the elements of the array.

Output Format:

- Print the rotated array as a single line of space-separated integers.

Constraints:

- `1 <= n <= 10^5`
- `0 <= k <= 10^9`
- `-1,000,000,000 <= array[i] <= 1,000,000,000`

Example:

Sample Input:

7 3
1 2 3 4 5 6 7

Sample Output:

```
5 6 7 1 2 3 4
```

Explanation:

NA

Public Test Cases:

#	INPUT	EXPECTED OUTPUT
1	7 3 1 2 3 4 5 6 7	5 6 7 1 2 3 4
2	5 0 10 20 30 40 50	10 20 30 40 50
3	6 6 -1 -100 3 99 10 20	-1 -100 3 99 10 20

Question 2:

Marks: +10 -0

Recursive String Collapse

You are given a string `S` consisting of lowercase English letters. Your task is to perform a series of "collapses". A collapse is defined as the removal of two adjacent identical characters. This process should be repeated until no more collapses can be performed.

For example, if the string is `abccbc`, the first collapse happens with `"cc"`, resulting in the string `"abbc"`. Now, in `"abbc"`, the characters `"bb"` are adjacent and identical, so they are collapsed. This results in the string `"ac"`. Since no more characters are adjacent and identical, `"ac"` is the final result.

The process is recursive in nature. After a collapse, the newly formed adjacent characters must also be checked for a potential collapse. For instance, in `abccba`, collapsing `"cc"` gives `abba`. Now, `"bb"` collapses to give `aa`. Finally, `"aa"` collapses to give an empty string.

Input Format:

- A single line containing the string `S`.

Output Format:

- A single line containing the final string after all possible collapses. If the final string is empty, print "Empty String".

Constraints:

- `1 <= length of S <= 1000`
- The string `S` will only contain lowercase English alphabetic characters (`a-z`).

Example:

Sample input:

geegsforgeeks

Sample Output:

```
sforgks
```

Explanation:

NA

Public Test Cases:

#	INPUT	EXPECTED OUTPUT
1	abccba	Empty String
2	mississippi	m
3	azxxza	Empty String

Question 3:

Marks: +10 -0

Cosmic Order Challenge

In the cosmic kingdom of Astrophoria, the Celestial Guardians protect the order of the universe. Each guardian possesses a unique identity represented by a string of characters, signifying their elemental properties. However, a celestial anomaly has caused the properties to become jumbled, and only the chosen guardians can restore order through a mystical process called the Cosmic Shift.
Each guardian can select up to k of their properties from the beginning of their identity string and shift any one of them to the end. But here lies the challenge: they are limited to performing no more than m shifts. The aim is to achieve a lexicographically smallest arrangement while adhering to a set of rules that complicate the process.

However, there's an additional requirement: Each time a guardian performs a shift, the string's properties must maintain a specific order based on their original positions, and they cannot shift the same character more than once in a single operation.

Constraints:

- 1 <= k <= s.length <= 1000
- 1 <= m <= 100
- s consists of lowercase English letters.

Example:

Example 1:

Input: ecbda 3 2 2
Output: baecd

Example 2:

Input: apple 3 2
Output: ppeal

Explanation:

Given a string s, two integers k (the number of properties available for selection) and m (the maximum number of shifts allowed), your task is to determine the smallest possible lexicographic arrangement of the string after performing no more than m shifts while following the ordering rule

Public Test Cases:

#	INPUT	EXPECTED OUTPUT
1	`helloworld 4 5`	`llwrldehoo`
2	`zxyw 3 1`	`zywx`

Question 4:

Marks: +10 -0

Quest for the Median Crown

In the mystical land of Numoria, two powerful kingdoms, Kingdom A and Kingdom B, have thrived in harmony, each renowned for their legendary warriors. The warriors of Kingdom A are represented by a sorted array of strengths, while those of Kingdom B are depicted by another sorted array of strengths. As the king of Numoria seeks to unite both kingdoms for a grand alliance, he must first determine the strength that symbolizes the true median warrior among the combined forces.
The challenge lies in finding the median strength, which is defined as the value that stands at the center of the merged array of warrior strengths. If the total number of warriors is even, the median is computed as the average of the two middle warriors' strengths.
To achieve this, you are tasked with devising an algorithm that can efficiently calculate the median strength of the united array, adhering to a time complexity of O(log(m+n))O(\log (m+n))O(log(m+n)), where mmm is the number of warriors in Kingdom A and n is the number in Kingdom B. This requirement underscores the need for a swift and strategic approach, befitting a noble strategist entrusted with such an important mission.

Constraints:

The strengths of the warriors are bound within the following limits:
- nums1.length == m
- nums2.length == n

- 0 ≤ m ≤ 1000
- 0 ≤ n ≤ 1000
- 1 ≤ m + n ≤ 2000
- -10^6 ≤ nums1[i], nums2[i] ≤ 10^6

Example:

Example 1:

Input: 1 3
2
Output: [1, 2, 3]
2.0

Example 2:

Input: 1 2
3 6
Output: [1, 2, 3, 6]
2.5

Explanation:

Your task is to develop an efficient algorithm that merges the strengths of both kingdoms' warriors, represented as two sorted arrays. Your solution must run in O(log(m+n))O(\log (m+n))O(log(m+n)) time complexity, ensuring a swift and cunning approach befitting a great strategist.
Your ultimate goal is to identify the strength that represents the true median warrior among the combined forces. If the total number of warriors is even, you will need to calculate the median as the average of the two middle strengths. Also, get the merged array along with the median in the output.

Public Test Cases:

#	INPUT	EXPECTED OUTPUT
1	0 0 0 0	[0, 0, 0, 0],0.0
2	1 5 9 2 3 7	[1, 2, 3, 5, 7, 9],4.0